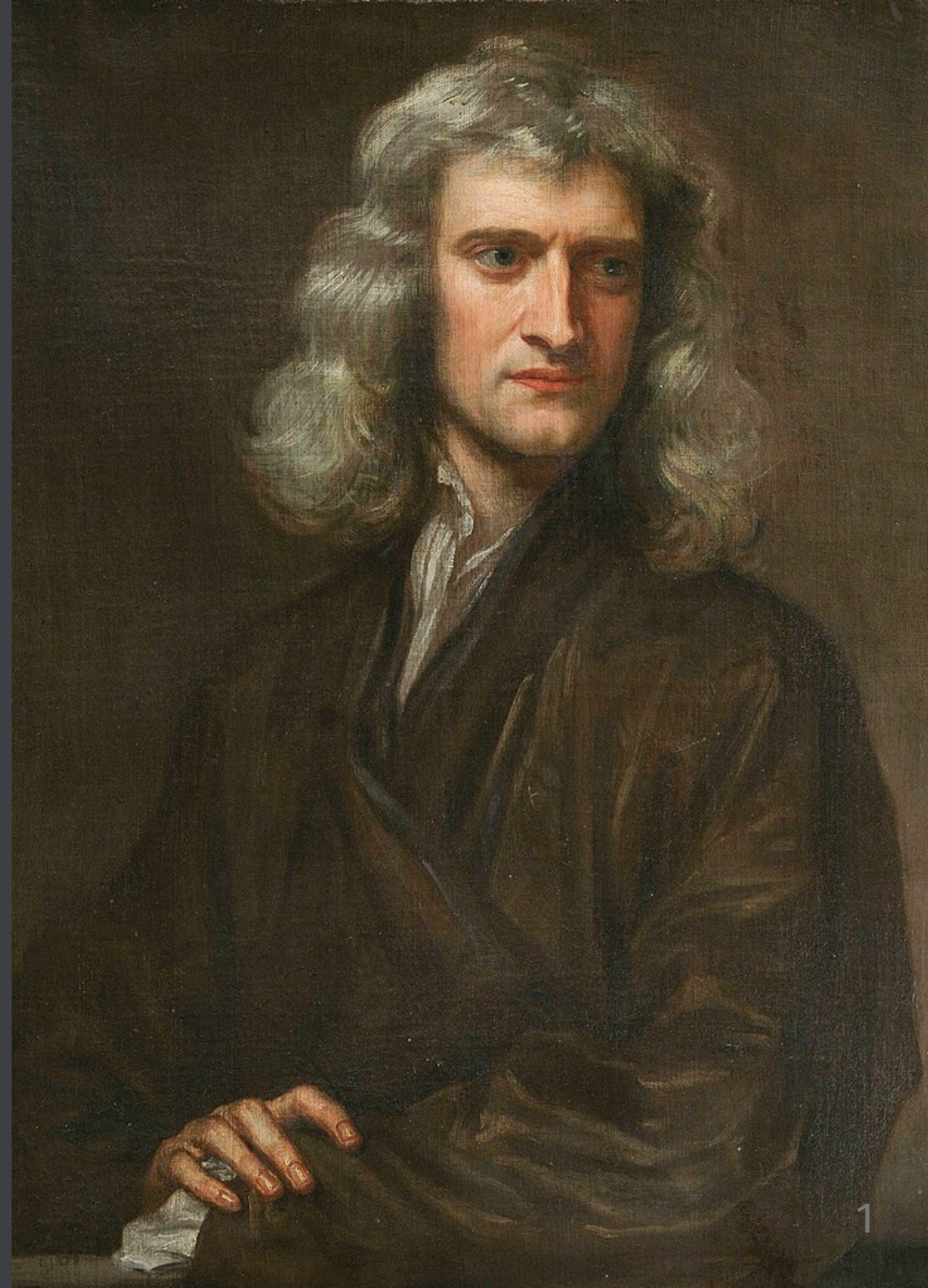


## Day 02 - Newton's Laws

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$$\mathbf{F}_{net} = ma$$



# Announcements

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- Homework 1 is due next Friday
- Help sessions will start next week
- Friday's class will include AI policy discussion
  - We will work Problem 1 together and start discussing Problem 6

# Goals for this week

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**Be able to answer the following questions.**

- What is Classical Mechanics?
- How can we formulate it?
- What are the essential physics models for single particles?
- What mathematics do we need to get started?

**Take 2 min to write down what comes to mind when asked:**

**What is "Classical" Physics?**

# Classical Mechanics

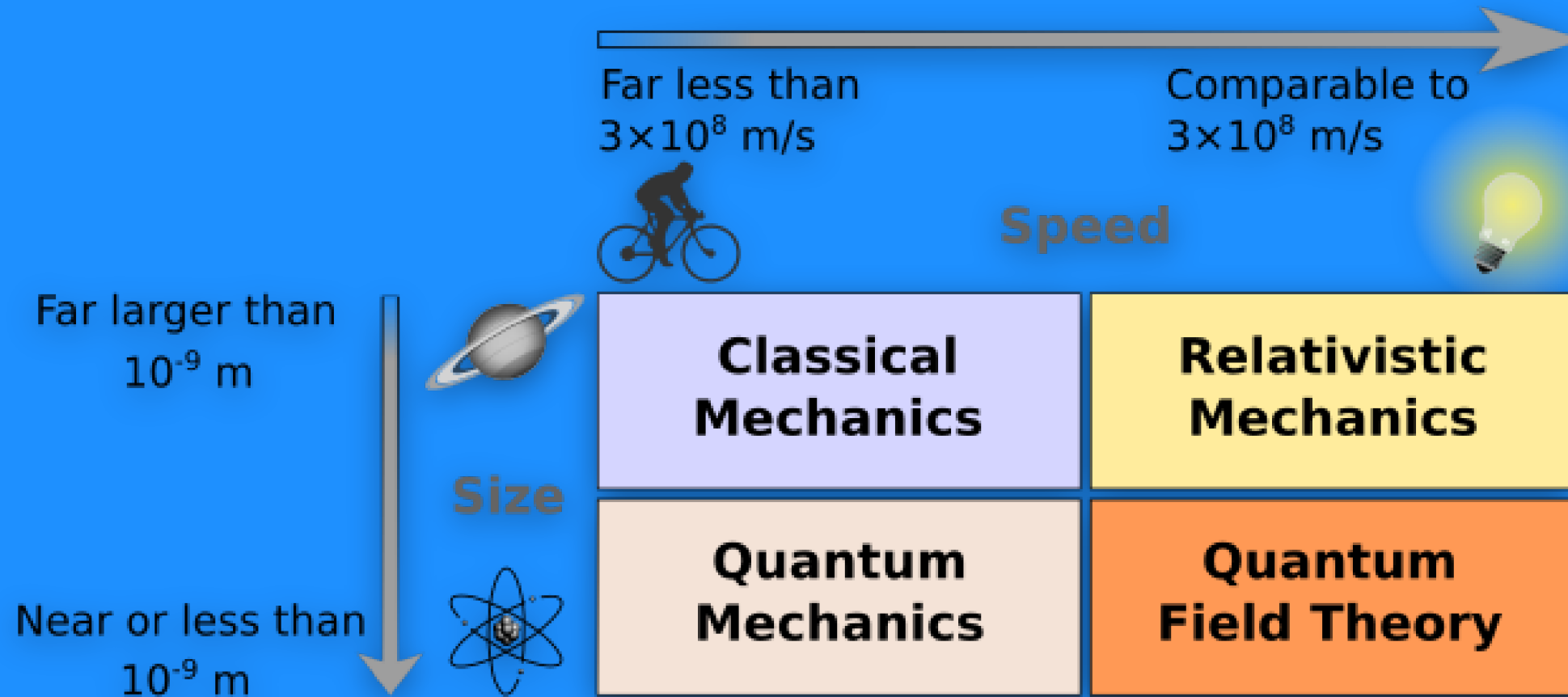
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## Modeling large, slow-moving objects

Newton's Laws are but one of a number of formulations:

- Lagrangian Mechanics
- Hamiltonian Mechanics
- Dynamical Systems Theory
- ...

# An Overview of Different Physics





# Classical Mechanics is still very relevant

Tiny Limbs and Long Bodies: Coordinating Lizard Locomotion

Research Lab



Source: <https://youtu.be/Qme07fA3Fj4>

## Think-Pair-Share

We used a tilted coordinate system ( $x - y$  plane) to analyze the motion of a block on an inclined plane. How can we check that we did the gravitational force decomposition correctly?

Recall:

- $F_{\text{gravity}, x} = mg \sin(\theta)$
- $F_{\text{gravity}, y} = mg \cos(\theta)$

Come up with at least two checks.



## Clicker Question 2-1

The formal definition of a Taylor series expansion around a point  $a$  is:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

This formula makes me feel:

1. Confident, I got this.
2. A little nervous, but I think I remember.
3. Uncomfortable, I don't remember this.
4. I have no idea what this is.

## Think-Pair-Share

We derived the following differential equation for the falling ball in one-dimension:

$$\frac{d^2y}{dt^2} = +g - \frac{b}{m} \frac{dy}{dt} - \frac{c}{m} \left( \frac{dy}{dt} \right)^2$$

Let's assume the turbulent drag term is negligible. Is there an anti-derivative of the right-hand side of this equation? If so, what is it?

$$\frac{dv}{dt} = +g - \frac{b}{m} v$$